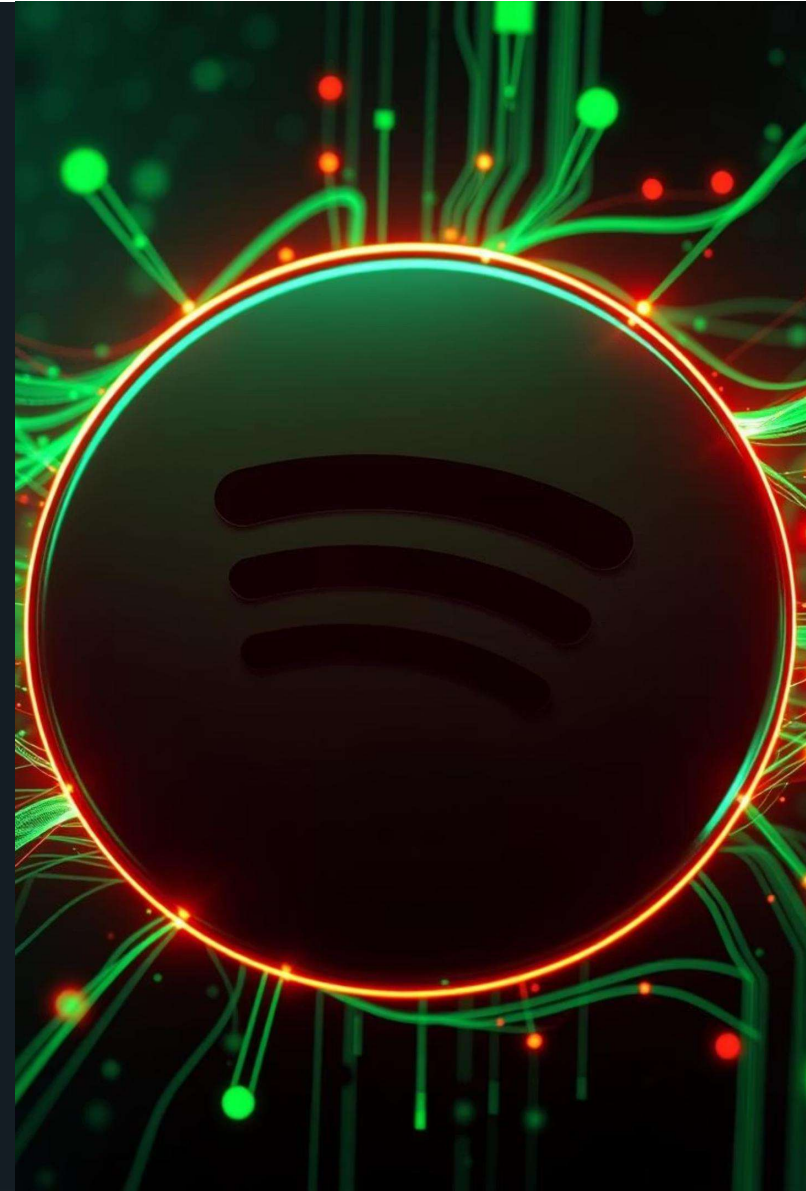


# Predicting Spotify User Churn

This project uses machine learning to predict whether a Spotify user will churn or stay, based on their listening behaviour and account details.



# Core Functionality

## Data Handling

Reads, processes, and cleans data from a Kaggle CSV dataset.

## Model Training

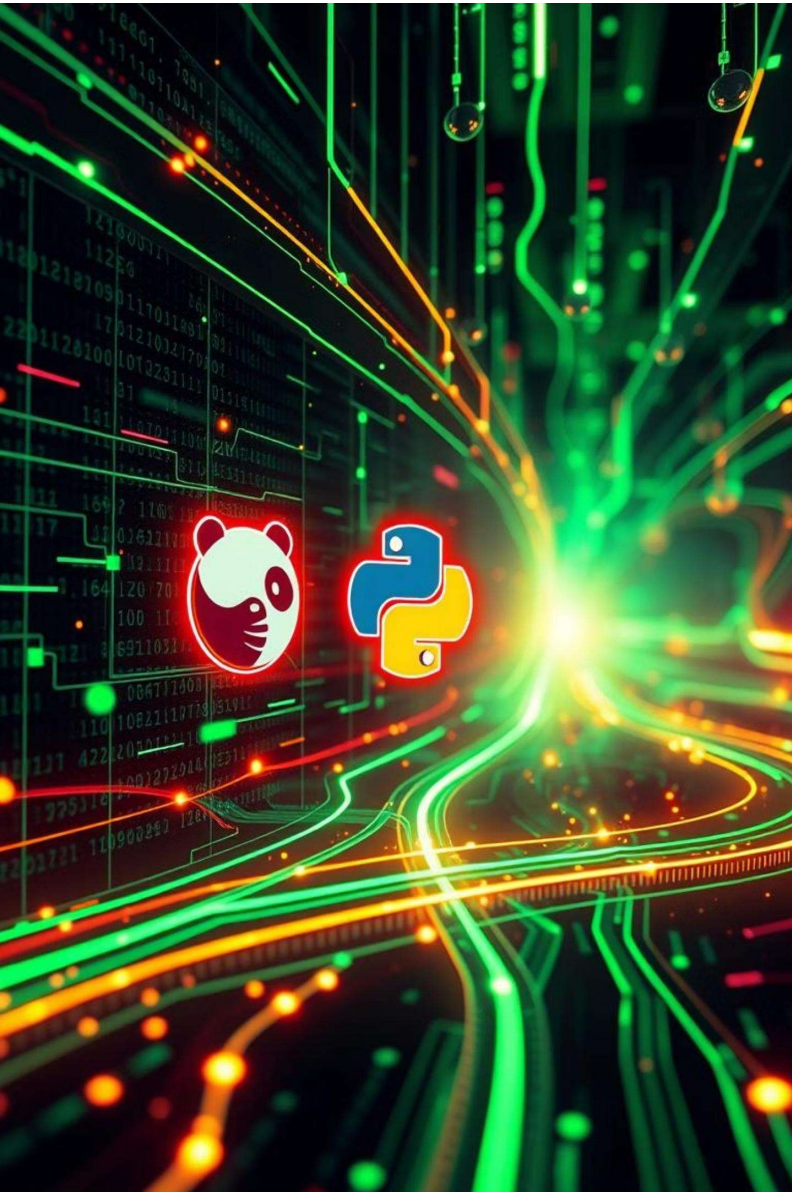
Trains a Random Forest model for accurate churn prediction.

## Model Persistence

Saves the trained model, scaler, and feature names.

## Interactive App

Provides a Gradio web app for real-time predictions.



# Tech Stack: Data Handling



**Pandas**

For loading and cleaning the dataset.



**NumPy**

For numerical computations.



# Tech Stack: Machine Learning



## Scikit-Learn

For preprocessing (StandardScaler) and model training (Random Forest Classifier).



## Joblib

To save and load the trained model, scaler, and feature names.



# Tech Stack: Interface & Deployment

A green circle containing a white code symbol (</>), positioned on a horizontal green line.

## Gradio

For building an interactive web app to make predictions.

A green circle containing a white cloud icon, positioned on a horizontal green line.

## Google Colab

For easy execution and sharing of the app.



# Dataset Source

## Kaggle

Spotify user churn dataset

Kaggle is a renowned online platform for data science and machine learning enthusiasts, hosting a vast collection of datasets and competitions. It serves as a valuable resource for aspiring and experienced data scientists to find, publish, and solve real-world data challenges.

Our project specifically utilizes a comprehensive Spotify user churn dataset sourced from Kaggle. This dataset contains a rich array of features related to user activity and demographics, such as subscription duration, daily listening hours, number of songs played, and interactions with various Spotify features. This detailed information is critical for training our machine learning model to accurately identify patterns that precede user churn.





# Input Fields on Web Interface

## User Profile

- User's Subscription Type
- Geographic Country
- Days Since Last Login

## User Listening Habits

- Average Daily Listening Minutes
- Number of Created Playlists
- User's Top Streamed Genre
- Average Daily Track Skips
- Number of Support Tickets Opened





# Example Prediction

## Input Data

- Subscription Type: Free
- Country: India
- Avg Daily Minutes: 45
- Number of Playlists: 5
- Top Genre: Pop
- Skips per Day: 2
- Support Tickets: 0
- Days Since Last Login: 5

## Predicted Outcome

User will likely stay (Probability:  
73.40%)



# How to Run on Google Colab

## 1 Open Colab

Open your Google Colab Notebook.

## 2 Paste Code

Copy and paste the provided project code.

## 3 Load Dataset

Load the provided dataset.

## 4 Run Cell

Run the cell and wait for a Gradio link.

## 5 Click Link

Click the link to access the form.

## 6 Predict

Fill details and click Predict to see the result.

